

SLIIT
SE3082 – Parallel Computing



Lab 7
Exercise 07

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Exercise 07: Summary Comparison Report

1. PROGRAM COMPARISON TABLE:

Program	Collectives Used	Memory Allocation	Manual Sum Loop?	Where is the final result?
1. Bcast	MPI_Bcast	Every process allocates the FULL array	Yes	Root only
2. Scatter	MPI_Scatter	Only root allocates full array, others allocate a chunk	Yes	Root only
3. Gather	MPI_Scatter, MPI_Gather	Only root allocates full array, others allocate a chunk	Yes	Root only
4. Reduce	MPI_Scatter, MPI_Reduce	Only root allocates full array, others allocate a chunk	No	Root only
5. Allreduce	MPI_Scatter, MPI_Allreduce	Only root allocates full array, others allocate a chunk	No	All processes
6. Scan	MPI_Scatter, MPI_Scan	Only root allocates full array, others allocate a chunk	No	Different prefix per rank

2. EXECUTION TIMES:

- sum_bcast: 2 procs = 0.0085s, 4 procs = 0.0121s, 8 procs = 0.0184s
- sum_scatter: 2 procs = 0.0042s, 4 procs = 0.0055s, 8 procs = 0.0076s
- sum_gather: 2 procs = 0.0038s, 4 procs = 0.0048s, 8 procs = 0.0069s
- sum_reduce: 2 procs = 0.0012s, 4 procs = 0.0015s, 8 procs = 0.0022s
- sum_allreduce: 2 procs = 0.0014s, 4 procs = 0.0018s, 8 procs = 0.0025s
- sum_scan: 2 procs = 0.0016s, 4 procs = 0.0021s, 8 procs = 0.0031s

FASTEST APPROACH: MPI_Reduce / MPI_Allreduce are generally the fastest.

WHY: They avoid the heavy manual for loops on the root process and use highly optimized internal MPI math trees to calculate the sums instead of sending raw arrays back and forth.

3. **THINKING QUESTION:**

Q: In what situation would you choose MPI_Scan over MPI_Allreduce?

A: You would use MPI_Scan when you need to calculate running totals (prefix sums) where sequence matters, rather than just one grand total. For example, if you are assigning global memory indexes to items processed by different computers in parallel. If Rank 0 processes 50 items and Rank 1 processes 30 items, MPI_Scan tells Rank 1 that its items should start at index 51, because it knows the exact total of everything that came before it. MPI_Allreduce only gives the grand total (80), which wouldn't help Rank 1 know where to start.